

1. Create a new user named `student1` and verify that the user has been created successfully.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo adduser student1
[sudo] password for pgcp-esd:
Adding user `student1' ...
Adding new group `student1' (1012) ...
Adding new user `student1' (1009) with group `student1' ...
Creating home directory `/home/student1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student1
Enter the new value, or press ENTER for the default
    Full Name []: student1
    Room Number []: 542357
    Work Phone []: 157143
    Home Phone []: 543525
    Other []: 523542
Is the information correct? [Y/n] y
```

```
alex:x:1002:1002:Alexa Kumari,107,1212,2121,1122:/home/alex:/bin/bash
Show Applications 1012:student1,542357,157143,543525,523542:/home/student1:/bin/bash
```

2. Create three users named `dev1`, `dev2`, and `dev3` and verify their accounts.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo adduser dev1
Adding user `dev1' ...
Adding new group `dev1' (1013) ...
Adding new user `dev1' (1010) with group `dev1' ...
Creating home directory `/home/dev1' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for dev1
Enter the new value, or press ENTER for the default
    Full Name []: dev1
    Room Number []: 56524
    Work Phone []: 524566
    Home Phone []: 256246
    Other []: 24654
Is the information correct? [Y/n] y
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo adduser dev2
Adding user `dev2' ...
Adding new group `dev2' (1014) ...
Adding new user `dev2' (1011) with group `dev2' ...
Creating home directory `/home/dev2' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for dev2
Enter the new value, or press ENTER for the default
    Full Name []: dev2
    Room Number []: 5654
    Work Phone []: 2654
    Home Phone []: 563456
    Other []: 2643653
Is the information correct? [Y/n] y
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo adduser dev3
Adding user `dev3' ...
Adding new group `dev3' (1015) ...
Adding new user `dev3' (1012) with group `dev3' ...
Creating home directory `/home/dev3' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for dev3
Enter the new value, or press ENTER for the default
    Full Name []: dev3
    Room Number []: 525625
    Work Phone []: 25245
    Home Phone []: 24526
    Other []: 25625
Is the information correct? [Y/n] y
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo cat /etc/passwd
```

```
dev1:x:1010:1013:dev1,56524,524566,256246,24654:/home/dev1:/bin/bash
dev2:x:1011:1014:dev2,5654,2654,563456,2643653:/home/dev2:/bin/bash
dev3:x:1012:1015:dev3,525625,25245,24526,25625:/home/dev3:/bin/bash
pgcp-esd@pgcpesd-Precision-3640-Tower:~$
```

3. Create a group named `developers` and verify its creation.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo groupadd developers
dev1:x:1013:
dev2:x:1014:
dev3:x:1015:
developers:x:1016:
pgcp-esd@pgcpesd-Precision-3640-Tower:~$
```

4. Add `dev1`, `dev2`, and `dev3` to the `developers` group.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo usermod -aG developers dev1
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo usermod -aG developers dev2
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ sudo usermod -aG developers dev3
pgcp-esd@pgcpesd-Precision-3640-Tower:~$
```

5. Display the groups to which `dev1` belongs.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ groups dev1
dev1 : dev1 developers
```

6. Create a directory named `project` and determine its current permissions.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ mkdir project
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ ls -l
```

```
drwxrwxr-x  2 pgcp-esd pgcp-esd 4096 Sep  2 15:36 project
```

7. Create a file named `project.txt` inside the `project` directory and determine its current permissions.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~$ cd project
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ touch project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ ls -l project.txt
-rw-rw-r-- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
```

8. Change the permissions of `project.txt` so that the owner has read, write, and execute permissions, the group has read and write permissions, and others have no permissions.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ sudo chmod a-rwx,u+rw,g+rw project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ ls -l project.txt
-rwxrw---- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$
```

9. Change the permissions of `project.txt` using octal notation so that the owner can read and write, the group can read only, and others have no access.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ sudo chmod 640 project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ ls -l project.txt
-rw-r----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$
```

10. Remove write permission for the group from `project.txt` without changing any other permissions.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$ ls -l project.txt
-rw-r----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/project$
```

11. Add execute permission for the owner of `project.txt` without changing the existing permissions of the group and others.

```
pgcp-esd@pc:~/project$ sudo chmod u+x project.txt
pgcp-esd@pc:~/project$ ls -l project.txt
-rwxr----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
pgcp-esd@pc:~/project$
```

12. Remove all permissions for others from every file in the `project` directory.

```
pgcp-esd@pc:~/project$ sudo chmod o-rwx project
pgcp-esd@pc:~/project$ ls -l project
total 0
-rwxr----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
```

13. Set permissions on the `project` directory so that the owner has full access, the group can read and enter the directory, and others have no access.

```
pgcp-esd@pc:~$ sudo chmod 740 project
pgcp-esd@pc:~$ ls -l project
total 0
-rwxr----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
```

14. Create a file named `report.txt` and use symbolic notation with `chmod` to give read permission to the owner and group while removing read permission from others.

```
pgcp-esd@pc:~/project$ touch report.txt
pgcp-esd@pc:~/project$ sudo chmod a-rwx,u+r,g+r report.txt
pgcp-esd@pc:~/project$ ls -l report.txt
-r--r----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:54 report.txt
```

15. Create a directory named `shared` and configure its permissions so that members of a group can create and modify files inside it but users outside the group cannot access it.

```
pgcp-esd@pc:~/project$ mkdir shared
pgcp-esd@pc:~/project$ sudo chmod 660 shared
pgcp-esd@pc:~/project$ ls -l shared
total 0
pgcp-esd@pc:~/project$ ls -l
total 4
-rwxr----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:39 project.txt
-r--r----- 1 pgcp-esd pgcp-esd 0 Sep  2 15:54 report.txt
drw-rw---- 2 pgcp-esd pgcp-esd 4096 Sep  2 15:56 shared
```

16. Create a group named `testers` and add `dev2` and `dev3` to it.

```
pgcp-esd@pc:~/project$ sudo groupadd testers
pgcp-esd@pc:~/project$ sudo usermod -aG testers dev2
pgcp-esd@pc:~/project$ sudo usermod -aG testers dev3
pgcp-esd@pc:~/project$ groups dev2
dev2 : dev2 developers testers
pgcp-esd@pc:~/project$ groups dev3
dev3 : dev3 developers testers
```

17. Remove `dev3` from the `testers` group without deleting the user account.

```
dev3 : dev3 developers testers
pgcp-esd@pc:~/project$ sudo gpasswd -d dev3 testers
Removing user dev3 from group testers
pgcp-esd@pc:~/project$
```

18. Change the primary group of `dev1` to `developers`.

```
pgcp-esd@pc:~/project$ sudo usermod -g developers dev1
pgcp-esd@pc:~/project$ groups dev1
dev1 : developers
```

19. Create a user named `manager` with `developers` as the user's primary group.

```
pgcp-esd@pc:~/project$ sudo useradd manager -m -s /bin/bash
pgcp-esd@pc:~/project$ sudo cat /etc/passwd
```

```
manager:x:1013:1018::/home/manager:/bin/bash
```

```
pgcp-esd@pc:~/project$ sudo usermod -g developers manager
pgcp-esd@pc:~/project$ groups manager
manager : developers admins
```

20. Display the UID, primary GID, and supplementary groups of `manager`.

```
pgcp-esd@pc:~/project$ id manager
uid=1013(manager) gid=1016(developers) groups=1016(developers),1019(admins)
```

21. Create a group named `admins` and add `manager` to it as a supplementary group.

```
pgcp-esd@pc:~/project$ sudo groupadd admins
pgcp-esd@pc:~/project$ sudo usermod -aG admins manager
pgcp-esd@pc:~/project$ groups manager
manager : developers admins
```

22. Display all members of the `developers` group. **Check with sir**

```
pgcp-esd@pc:~/project$ getent group developers
developers:x:1016:dev1,dev2,dev3
```

23. Display all members of the `admins` group.

```
pgcp-esd@pc:~/project$ getent group admins
admins:x:1019:manager
```

24. Create a user named `operators` and verify the user's UID, GID, home directory, and login shell.

```
pgcp-esd@pc:~/project$ sudo useradd operators -m -s /bin/bash
pgcp-esd@pc:~/project$ id operators
uid=1014(operators) gid=1020(operators) groups=1020(operator
s)
```

```
manager:x:1013:1018:~/none/manager:/bin/bash
operators:x:1014:1020:~/home/operators:/bin/bash
```

25. Create a user named `intern` with a specific home directory and verify that the home directory has been created.

```
pgcp-esd@pc:~/project$ sudo useradd intern -m -s /bin/bash
pgcp-esd@pc:~/project$ sudo cat /etc/passwd
```

```
operator:x:1014:1020:~/none/operator:/bin/bash
intern:x:1015:1021:~/home/intern:/bin/bash
```

26. Change the login shell of `intern` and verify the change.

```
pgcp-esd@pc:~/project$ sudo usermod -s /bin/sh intern
pgcp-esd@pc:~/project$ sudo cat /etc/passwd | tail -1
intern:x:1015:1021:~/home/intern:/bin/sh
```

27. Set a password for `student1` and verify that the user can log in using `su`.

```
pgcp-esd@pc:~/project$ sudo passwd student1
New password:
Retype new password:
passwd: password updated successfully
```

```
pgcp-esd@pc:~/project$ su student1
Password:
student1@pc:/home/pgcp-esd/project$
```

28. Create a file named `admin.txt` and determine its default permissions.

```
pgcp-esd@pc:~/project$ touch admin.txt
pgcp-esd@pc:~/project$ ls -l admin.txt
-rw-rw-r-- 1 pgcp-esd pgcp-esd 0 Sep  2 17:26 admin.txt
pgcp-esd@pc:~/project$
```

29. Create a directory named `teamshare` and configure its permissions so that only the owner and group members can access it.

```
pgcp-esd@pc:~/project$ mkdir teamshare
pgcp-esd@pc:~/project$ sudo chmod o-rwx teamshare
pgcp-esd@pc:~/project$ ls -l teamshare
drwxrwx--- 2 pgcp-esd pgcp-esd 4096 Sep  2 17:27 teamshare
```

30. Configure `teamshare` so that files created inside it cannot be accessed by users outside the group.

```
pgcp-esd@pc:~/project$ sudo setfacl -d -m o::--- teamshare
pgcp-esd@pc:~/project$ getfacl teamshare
# file: teamshare
# owner: pgcp-esd
# group: pgcp-esd
user::rwx
group::rwx
other::---
default:user::rwx
default:group::rwx
default:other::---
```

31. Create a file as `dev1`, switch to `dev2`, and determine whether `dev2` can read, modify, or delete the file based on its permissions.

```
pgcp-esd@pc:~/project$ su dev1
Password:
dev1@pc:~/project$ cd ~
dev1@pc:~$ touch File.txt
dev1@pc:~$ su dev2
Password:
dev2@pc:~/dev1$ cd ~
dev2@pc:~$ cd -
/home/dev1
dev2@pc:~/dev1$ vim File.txt
```

```
dev1@pc:~/project$ cd ~
dev1@pc:~$ ls -l File.txt
-rw-r--r-- 1 dev1 developers 25 Sep  2 17:46 File.txt
dev1@pc:~$
```

```
this is created by dev1.
changes by dev2
```

```
~
~
E45: 'readonly' option is set
(add ! to override)
Press ENTER or type command t
o continue
```

32. Create a directory accessible only by `dev1` and test whether `dev2` can access it.

```
dev1@pc:~$ mkdir DIR1 && chmod g-rwx,o-rwx DIR1
```

```
dev1@pc:~$ getfacl DIR1
# file: DIR1
# owner: dev1
# group: developers
user::rwx
group::-
other::-
```

33. Create a directory that allows group members to create files but prevents them from deleting files created by other group members.

34. Create a file with permissions `640` and test access to the file using users with different group memberships.

```
pgcp-esd@pc:~$ echo "This is a text file" > newFile.txt
pgcp-esd@pc:~$ sudo chmod 640 newFile.txt
[sudo] password for pgcp-esd:
pgcp-esd@pc:~$ su dev1
Password:
dev1@pc:/home/pgcp-esd$ vim newFile.txt
```

```
~
"newFile.txt" [Permission Denied]
```

35. Create a file with permissions `754` and determine which operations are possible for the owner, group members, and others.

```
pgcp-esd@pc:~$ sudo chmod 754 newFile.txt
pgcp-esd@pc:~$ ls -l newFile.txt
-rwxr-xr-- 1 pgcp-esd pgcp-esd 20 Sep  3 00:28 newFile.txt
pgcp-esd@pc:~$ getfacl newFile.txt
# file: newFile.txt
# owner: pgcp-esd
# group: pgcp-esd
user::rwx
group::r-x
other::r--
```

36. Create a file with permissions ``600`` and test whether another user can read or modify it.

```
pgcp-esd@pc:~$ sudo chmod 600 newFile.txt
pgcp-esd@pc:~$ su dev1
Password:
dev1@pc: /home/pgcp-esd$ vim newFile.txt
```

```
"newFile.txt" [Permission Denied]
```

37. Create a file with permissions `777` and explain, through practical testing, what access is provided to different users.

```
pgcp-esd@pc:~$ mkdir permission
pgcp-esd@pc:~$ touch permission/share.txt
pgcp-esd@pc:~$ sudo chmod 777 permission/share.txt
[sudo] password for pgcp-esd:
pgcp-esd@pc:~$ sudo chmod a+r pgcp-esd
chmod: cannot access 'pgcp-esd': No such file or directory
pgcp-esd@pc:~$ sudo chmod a+r permission
pgcp-esd@pc:~$ cd permission
pgcp-esd@pc:~/permission$ su dev1
Password:
dev1@pc: /home/pgcp-esd/permission$ vim share.txt
dev1@pc: /home/pgcp-esd/permission$
```

[illegible]

38. Verify the current user's sudo privileges using the appropriate command.

```
dev1@pc:/home/pgcp-esd/permissions$ cat share.txt
dev1@pc:/home/pgcp-esd/permissions$ sudo cat /etc/sudoers
[sudo] password for dev1:
dev1 is not in the sudoers file. This incident will be reported.
dev1@pc:/home/pgcp-esd/permissions$ exit
exit
pgcp-esd@pc:~/permissions$ sudo cat /etc/sudoers
#
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

39. Execute a command as the root user using `sudo` and verify the effective user identity.

```
pgcp-esd@pc:~/permission$ sudo whoami  
root
```

40. Use `sudo` to execute a command that requires administrative privileges.

```
pgcp-esd@pc:~$ sudo echo "testing sudo privileges" > test.txt  
[sudo] password for pgcp-esd:  
pgcp-esd@pc:~$
```

41. Create a user named `operator` and configure `sudo` so that the user can execute a specific administrative command.

```
pgcp-esd@pc:~$ sudo usermod -s /bin/bash operator  
pgcp-esd@pc:~$ sudo usermod -aG sudo dev1  
pgcp-esd@pc:~$ sudo visudo  
visudo: /etc/sudoers.tmp unchanged  
pgcp-esd@pc:~$ su dev1  
Password:  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
dev1@pc:/home/pgcp-esd$ sudo touch hello/txt  
[sudo] password for dev1:  
touch: cannot touch 'hello/txt': No such file or directory  
dev1@pc:/home/pgcp-esd$ sudo touch hello.txt  
dev1@pc:/home/pgcp-esd$
```

42. Configure `sudo` so that members of the `admins` group can execute administrative commands.

```
pgcp-esd@pc:~$ sudo visudo  
%admin ALL=(ALL) ALL  
%admins ALL=(ALL:ALL) ALL
```

43. Switch from the current user to `manager` using `su` and verify the effective user identity.

```
pgcp-esd@pc:~$ su manager
Password:
manager@pc:/home/pgcp-esd$ whoami
manager
```

44. Use `su` to switch to `manager` while loading the user's login environment.

```
pgcp-esd@pc:~$ su manager -l
Password:
manager@pc:~$ echo $HOME
/home/manager
manager@pc:~$
```

45. Use `su` to execute a single command as `manager` without starting an interactive shell.

```
pgcp-esd@pc:~$ su manager -c 'echo 'Hello''
Password:
Hello
pgcp-esd@pc:~$
```

46. Switch to the root account using `su` and verify the current user.

```
pgcp-esd@pc:~$ sudo -i
root@pc:~# whoami
root
root@pc:~#
```

47. Compare the output of `id`, `whoami`, and `groups` before and after using `su` to switch users.

```
pgcp-esd@pc:~$ id
uid=1000(pgcp-esd) gid=1000(pgcp-esd) groups=1000(pgcp-esd),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),122(lpadmin),135(lxd),136(sambashare)
pgcp-esd@pc:~$ whoami
pgcp-esd
pgcp-esd@pc:~$ groups pgcp-esd
pgcp-esd : pgcp-esd adm cdrom sudo dip plugdev lpadmin lxd sambashare
pgcp-esd@pc:~$ su manager
Password:
manager@pc:/home/pgcp-esd$ id
uid=1013(manager) gid=1016(developers) groups=1016(developers),1019(admins)
manager@pc:/home/pgcp-esd$ whoami
manager
manager@pc:/home/pgcp-esd$ groups manager
manager : developers admins
manager@pc:/home/pgcp-esd$
```

48. Create a file named `aclfile.txt` and use `getfacl` to display its existing ACL.

```
pgcp-esd@pc:~$ touch aclfile.txt
pgcp-esd@pc:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: pgcp-esd
# group: pgcp-esd
user::rw-
group::rw-
other::r--
```

49. Use `setfacl` to give `dev2` read permission on `aclfile.txt` without changing the normal file permissions, then verify the ACL using `getfacl`.

```
pgcp-esd@pc:~$ setfacl -m u:dev2:r aclfile.txt
pgcp-esd@pc:~$ getfacl aclfile.txt
# file: aclfile.txt
# owner: pgcp-esd
# group: pgcp-esd
user::rw-
user:dev2:r--
group::rw-
mask::rw-
other::r--
```

50. Create a directory named `aclshare` and configure ACLs so that `dev1` has full access, `dev2` has read and execute access, and `dev3` has no access; verify the configuration by testing access with each user.

```
pgcp-esd@pc:~$ mkdir aclshare
pgcp-esd@pc:~$ setfacl -m u:dev1:rw,x,u:dev2:r-x,u:dev3:--- aclshare
pgcp-esd@pc:~$ su dev1
Password:
dev1@pc:/home/pgcp-esd$ cd aclshare
bash: cd: aclshare: Permission denied
dev1@pc:/home/pgcp-esd$ exit
exit
pgcp-esd@pc:~$ sudo chmod o+x ~
pgcp-esd@pc:~$ su dev1
Password:
dev1@pc:/home/pgcp-esd$ cd aclshare
dev1@pc:/home/pgcp-esd/aclshare$ echo "Test file content" test.txt
Test file content test.txt
dev1@pc:/home/pgcp-esd/aclshare$
```

```
dev1@pc:/home/pgcp-esd$ su dev2
Password:
dev2@pc:/home/pgcp-esd$ cd aclshare
dev2@pc:/home/pgcp-esd/aclshare$ ls
dev2@pc:/home/pgcp-esd/aclshare$ touchtestfile2.txt
touchtestfile2.txt: command not found
dev2@pc:/home/pgcp-esd/aclshare$ touch testfile2.txt
touch: cannot touch 'testfile2.txt': Permission denied
dev2@pc:/home/pgcp-esd/aclshare$
```

```
touch: cannot touch 'testfile2.txt': Permission denied
dev2@pc:/home/pgcp-esd/aclshare$ su dev3
Password:
dev3@pc:/home/pgcp-esd/aclshare$ ls
ls: cannot open directory '.': Permission denied
dev3@pc:/home/pgcp-esd/aclshare$
```